Technical Requirements Document

Enterprise IT Support Agentic RAG Copilot

Document Version	1.0.0 (Production Architecture)
Technology Stack	LangGraph, FastAPI, Pinecone, Groq (gpt-oss-120b), Tavily, SQLite
Author / Lead Engineer	Bittu Sharma (FDE Systems Architect & AI Engineer)
Status	Architecture Verified & Tested (38 Unit Tests Passed)
Vector Engine	Pinecone Serverless (AWS us-east-1, Cosine, 384-dim)
Embedding Model	sentence-transformers/all-MiniLM-L6-v2 (Normalized)

1. System Architecture & Component Design

The system decouples user presentation from the agentic control plane. FastAPI provides high-throughput asynchronous HTTP endpoints, while LangGraph manages stateful graph traversal with cyclic loop control.

Layer	Technology	Technical Responsibility
Presentation	Vanilla HTML5 / CSS3 / ES6+	Dual-tab state, real-time KPI progress bars, trace visualization, no external bloated dependencies.
API Gateway	FastAPI 0.116 / Uvicorn	REST endpoints (/api/chat, /api/health, /api/ingest), header auth, CORS, async request dispatching.
Graph Engine	LangGraph 1.1.10	StateGraph execution: route -> retrieve -> grade -> web fallback -> retry -> answer generation.
Reasoning LLM	Groq (openai/gpt-oss-120b)	Ultra-fast sub-second inference for routing, grading, query rewriting, and grounded response synthesis.
Vector Index	Pinecone Serverless (v7.3)	Enterprise namespace partitioning, 384-dimensional cosine index, dynamic index lifecycle management.
Audit Persistence	SQLite 3 (WAL mode)	Local zero-latency write of query timestamps, routed sources, and JSON-serialized decision traces.

2. LangGraph State Machine & Node Flow

The shared agent state is defined via typing.TypedDict with explicit schemas for decisions and grades:

- **AgentState Schema:** question, current_query, kb_docs (List[Document]), web_results (str), kb_grade (Literal['good','weak']), web_grade, answer, source_used, retry_count, trace (List[str]), citations (List[Dict]).
- **Conditional Branch 1 (Router):** route_after_router inspects source_used; dispatches to retrieve_kb or direct_answer.
- **Conditional Branch 2 (KB Grade):** after_kb routes to generate_from_kb if 'good', else searches external web via Tavily.

• **Conditional Branch 3 (Web Grade & Loop Guard):** after_web routes to generate_from_web if 'good'. If 'weak' and retry_count < max_retries (1), it rewrites the query and retries KB retrieval; otherwise terminates at insufficient.

SAFETY LOOP GUARD: Deterministic Loop Prevention: retry_count is incremented on every rewrite_query execution. When retry_count reaches settings.max_retries (1), the state machine halts with an actionable fallback message to prevent runaway API spend.

3. REST API Specifications

POST /api/chat

```
Request: { question: str (2-3000 chars) }
Response: { answer, source_used, trace, citations, rewritten_query, confidence_score,
groundedness_score }
```

GET /api/health

```
Response: { status: 'ok', service: 'Enterprise IT Support Agentic RAG Copilot' }
```

POST /api/ingest

```
Headers: X-Admin-Key:
Body: Multipart form-data with file (.pdf, .txt, .md, .docx)
Response: { message, file, chunks, ids_created }
```